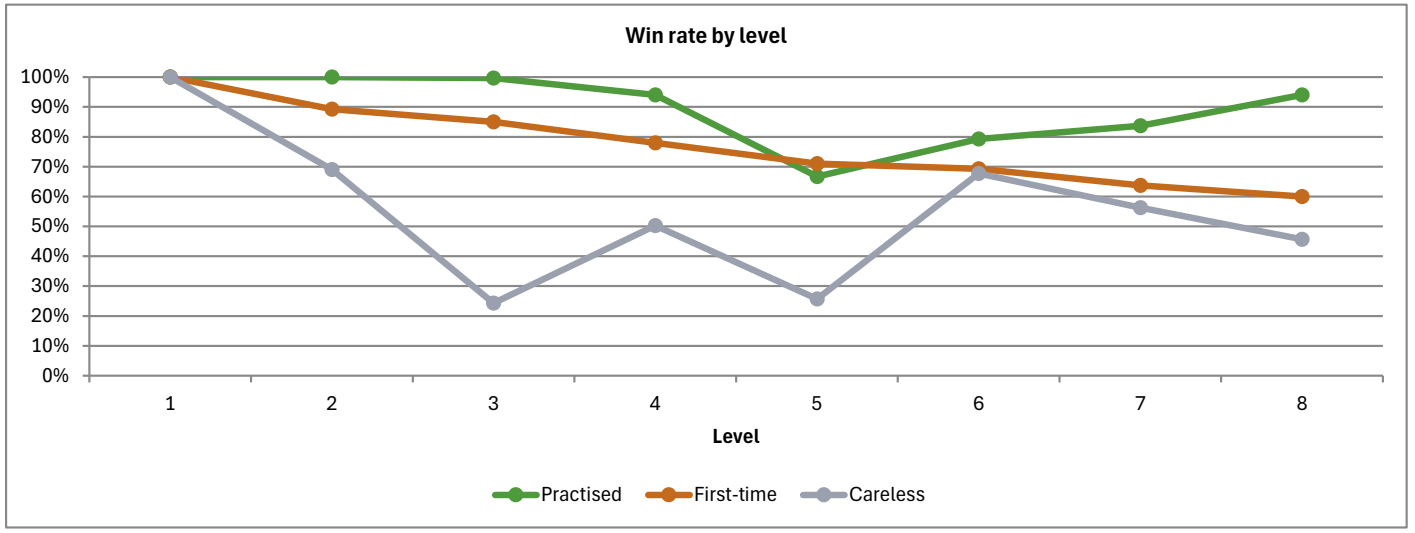
Bonemarch: balance and economy

The eight hand-made levels (first drafts, second pass). Bots play every level on the game's own rules (Bonemarch.Core); 300 battles a cell. Tools/sim/run.ps1 report regenerates the data.

Three bots stand in for kinds of player. Careless: any card, builds nearest the gate (the floor). First-time: takes the card that adds the most damage, heeds the intro's warning about flyers, builds nearest the gate. Practised: plans for the level's threats (flyers, armour, speed, crowds, healers), keeps its towers varied and builds where they cover the most road. All three build every tower they draft, spend embers on upgrades as soon as they can, and pick up 60% of the crystals. Bots never misread the map or panic: these numbers guide the tuning, they do not forecast real players.

Level	Name	First-time target	First-time	Practised target	Practised	Careless ceiling	Careless	Lives left (first-time wins)	Minutes of fighting (first-time)
1	Mossgate Meadow	95-100%	100%	95%+	100%	-	100%	18.5	2.5
2	Shieldwall Brook	85-95%	89%	95%+	100%	50%	69%	17.2	7.0
3	Ember Hollow	80-92%	85%	90%+	100%	45%	24%	12.9	7.7
4	Brute Ridge	75-88%	78%	90%+	94%	45%	50%	15.1	7.0
5	Wraith Fen	65-80%	71%	90%+	67%	45%	26%	16.3	7.8
6	Necromancer's Grove	62-75%	69%	85%+	79%	45%	68%	13.0	9.8
7	Giant's Steps	60-72%	64%	85%+	84%	45%	56%	9.3	13.5
8	The Old Wood	58-70%	60%	85%+	94%	40%	46%	10.7	13.5

Green: within 3 points of the target; red: outside it. The first-time player sits in its band on every level, falling from a sure tutorial to three wins in five. The practised player misses its floor on levels 5 and 6, and the careless player is over its ceiling on levels 2, 4 and 6 to 8: on the drafts, reading the cards and the map is not yet worth enough late on. LEVELS.md has the notes for the redesign.



Tuning

Every number the balance pass set. Blue cells are inputs; the tables below follow them. Copy changes into Bonemarch.Core/Tuning.cs and rerun the bots.

Input	Field	Value
Lives at the start	startLives	**20**
Embers at the start, level 1	startEmbers	**15**
... more for each later level	startEmbersPerLevel	**5**
Ruined pad: embers to clear	ruinCost	**40**
High ground: extra range	highGround	**0.25**
Marsh: share of walking speed	marshSpeed	**0.6**
Three stars: lives left	star3Lives	**18**
Two stars: lives left	star2Lives	**10**
Seconds between spawns	spawnGap	**0.9**
... between Runners	runnerGap	**0.5**
Health growth per wave	hpPerWave	**0.25**
Kill embers multiplier	killEmbers	**1**
Wave-clear embers: base	waveBonus	**8**
... per wave	waveBonusPerWave	**2**
Ember Surge: embers up front	surgeEmbers	**30**
Ember Surge: kill embers times	surgeMultiplier	**2**
Crystal value: base (+ level number)	crystalBase	**6**
Seconds between extra crystals	crystalInterval	**60**
Upgrade to level 2	upgradeCost2	**30**
Upgrade to level 3	upgradeCost3	**60**

Per level

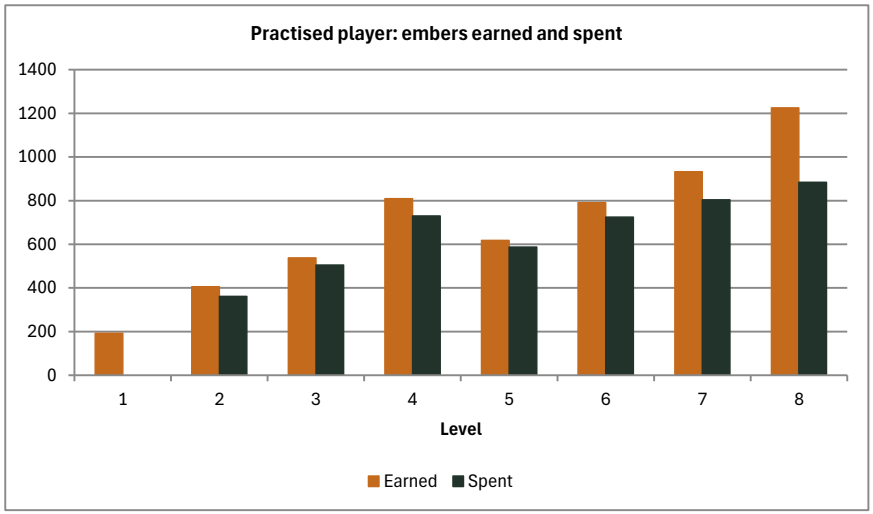
Level	Health multiplier (input)	Waves	Enemies in the level	Minion health, wave 1	Minion health, last wave	Embers at the start	Wave-clear embers in the level	Crystal value	One tower to level 3
1	**3**	6	58	60	135	15	78	7	90
2	**5.41**	8	100	108	298	20	120	8	90
3	**8.84**	8	110	177	486	25	120	9	90
4	**6.53**	9	137	131	392	30	144	10	90
5	**7.14**	9	135	143	428	35	144	11	90
6	**2.09**	9	148	42	125	40	144	12	90
7	**2.17**	9	121	43	130	45	144	13	90
8	**2.46**	10	205	49	160	50	170	14	90

Health multiplier: each level's enemies have their Defs health times this (a Minion has 20), growing by the per-wave rate within the level. run.ps1 fit set each level's value for the first-time player's target; level 1 was eased by hand to 3. Levels 6 to 8 sit lower because their waves bring more enemies, and tougher ones (Necromancers, Bone Giants). A level may set its own starting embers in Levels.txt; the table shows the default.

Embers

Average per battle. Earned: kills, wave clears, the Ember Surge and crystals. Spent: upgrades and ruins. Left: unspent at the end.

Player	Level	Earned	Spent	Left	Towers	Upgrades	Crystals taken	Crystals seen	Minutes
Careless	1	191	0	206	6.0	0.0	0	0	2.8
Careless	2	328	303	45	7.8	8.8	0	0	7.4
Careless	3	239	224	40	4.7	6.2	9	15.1	4.8
Careless	4	562	456	136	5.9	10.8	12.8	21.6	6.3
Careless	5	383	348	70	5.2	8.6	11.1	18.5	5.6
Careless	6	756	587	210	7.0	13.6	18	30.3	10.0
Careless	7	830	632	243	6.4	12.5	19.6	32.7	13.4
Careless	8	1054	657	447	7.5	15.0	21.5	36	13.5
First-time	1	191	0	206	6.0	0.0	0	0	2.5
First-time	2	386	351	55	8.0	9.8	0	0	7.0
First-time	3	488	464	49	8.8	12.5	15.3	25.5	7.7
First-time	4	729	661	98	8.8	15.8	15.2	25.4	7.0
First-time	5	653	585	103	8.1	14.2	15.8	26.3	7.8
First-time	6	763	684	119	8.5	16.2	18.1	30.2	9.8
First-time	7	874	755	164	7.6	14.7	20.1	33.5	13.5
First-time	8	1106	797	359	9.2	18.2	21.9	36.4	13.5
Practised	1	192	0	207	6.0	0.0	0	0	2.4
Practised	2	406	361	65	8.0	10.0	0	0	6.4
Practised	3	538	505	57	8.5	12.9	15	25	6.5
Practised	4	809	730	110	8.9	16.8	15.9	26.4	7.0
Practised	5	618	587	65	8.1	14.0	15.5	25.8	7.5
Practised	6	791	725	106	8.8	16.6	18.5	30.9	9.7
Practised	7	933	804	174	7.9	15.3	20.6	34.4	13.3
Practised	8	1225	884	391	9.9	19.8	21.4	35.5	11.0



Embers stay short through level 7: the first-time player ends levels 2 to 7 with 50 to 165 left, one or two upgrades' worth. Level 8 leaves about 360 unspent. The hand-made levels take 2.5 to 13.5 minutes of fighting (first-time player), where the generated maps of phase 3 took up to 37, which was where the unspent embers piled up.

Where battles are lost

Share of all battles lost in each wave, by level (300 battles each). An early wall feels unfair; a loss in the last waves feels earned.

First-time

Level	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Wave 6	Wave 7	Wave 8	Wave 9	Wave 10	Win
1	0%	0%	0%	0%	0%	0%					100%
2	0%	0%	0%	0%	0%	0%	2%	8%			89%
3	0%	0%	4%	0%	8%	1%	0%	1%			85%
4	0%	2%	9%	0%	10%	1%	0%	0%	0%		78%
5	0%	0%	15%	0%	8%	2%	0%	1%	2%		71%
6	0%	0%	0%	0%	12%	0%	0%	12%	6%		69%
7	0%	0%	7%	0%	1%	0%	7%	0%	22%		64%
8	0%	0%	0%	0%	0%	5%	0%	25%	0%	9%	60%

Practised

Level	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Wave 6	Wave 7	Wave 8	Wave 9	Wave 10	Win
1	0%	0%	0%	0%	0%	0%					100%
2	0%	0%	0%	0%	0%	0%	0%	0%			100%
3	0%	0%	0%	0%	0%	0%	0%	0%			100%
4	0%	0%	3%	0%	2%	0%	0%	0%	0%		94%
5	0%	0%	11%	1%	14%	4%	0%	1%	2%		67%
6	0%	0%	0%	0%	5%	1%	0%	11%	5%		79%
7	0%	0%	1%	0%	0%	0%	3%	0%	12%		84%
8	0%	0%	0%	0%	0%	2%	0%	1%	0%	3%	94%

Careless

Level	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Wave 6	Wave 7	Wave 8	Wave 9	Wave 10	Win
1	0%	0%	0%	0%	0%	0%					100%
2	0%	0%	4%	0%	6%	4%	10%	7%			69%
3	0%	18%	39%	3%	13%	2%	0%	1%			24%
4	0%	4%	29%	0%	10%	3%	0%	1%	2%		50%
5	0%	3%	44%	2%	11%	7%	0%	4%	3%		26%
6	0%	0%	0%	0%	12%	2%	0%	10%	8%		68%
7	0%	0%	9%	0%	3%	0%	11%	0%	20%		56%
8	0%	0%	0%	0%	1%	8%	1%	30%	0%	16%	46%

The first-time player's losses sit in the second half of most levels. Two walls stand out: level 5's third wave (15% of battles end there) and level 7's last, the Bone Giants (22%), meant as a set piece.

What the cards are worth

Change in win rate, in points. Differences under about 3 points are noise.

Boons: the first-time player takes it the first time it is offered and of use (levels 4 to 8)

Boon	Change
Watchtower Lens	2
Steady Hands	1
Dry Kindling	0
Fletcher's Oath	0
Hearthstone	0
Black Powder	-1
Deep Frost	-1
Guild Discount	-2
Copper Rod	-2
Ember Cache	-3

Towers: the practised player never drafts it (levels 3 to 8 that offer it)

Tower	Share of tower picks	Win rate with it	Change without it
Storm Tower	33%	81%	-10
Frost Tower	25%	86%	-10
Cannon Tower	32%	86%	-8
Flame Tower	17%	84%	-5
Arrow Tower	6%	86%	-1

The Ember Surge: win rate with and without it

Player	Level	With	Without	Worth
Careless	2	69%	40%	29
Careless	3	24%	13%	12
Careless	4	50%	25%	26
Careless	5	26%	9%	17
Careless	6	68%	60%	8
Careless	7	56%	40%	16
Careless	8	46%	45%	0
First-time	2	89%	80%	10
First-time	3	85%	72%	13
First-time	4	78%	60%	18
First-time	5	71%	61%	10
First-time	6	69%	62%	7
First-time	7	64%	50%	13
First-time	8	60%	52%	8
Practised	2	100%	99%	1
Practised	3	100%	99%	1
Practised	4	94%	82%	12
Practised	5	67%	51%	15
Practised	6	79%	73%	6
Practised	7	84%	64%	20
Practised	8	94%	91%	3

On the hand-made levels the Storm, Frost and Cannon Towers are each worth 8 to 10 points to the practised player and the Flame Tower 5; the Arrow, 6% of its picks, only 1. Boons sit between -3 and +2 points whatever their size: a boon replaces a tower card, and a new tower's fresh stretch of road is worth as much as a percentage on the towers already built. Making boons matter is a design question, set out in BALANCE.md. The Ember Surge is worth 10 to 18 points to the first-time player on levels 2 to 5, where the new enemies arrive.